2. TRD - How we build it

TRD means Technical Requirements Document. It lists tools and technical rules.

Tech stack (simple)

- Frontend: Next.js (website + app screens)
- Backend: Node.js (APIs and jobs)
- Database: Firebase Firestore
- Login: Firebase Auth + Google/Gmail
- Files: Firebase Storage (images, videos, ZIP)
- Email: Resend
- Hosting: Vercel
- Code: Git / GitHub
- Domain: DNS A-record points to Vercel

Logging (for marks)

Every important action writes a clear log: time, module name, user/campaign id, success or fail. Modules: auth, transcribe, clipping, plans.

Errors

- Show simple messages to the user
- Allow retry when possible
- Do not show scary technical stack traces
- Bad URL -> offer manual upload
- Plan limit -> show upgrade option

Unit testing

We write small tests for login helpers, plan limits, product field cleaning, and error codes. Tests run before deploy. Logging + errors + tests = quality section (about 7 marks).

Hosting and domain

App runs on Vercel. Custom domain uses DNS A-record / CNAME as Vercel shows. Emails go through Resend.